

Symbols and Notation

Bold symbols ($\mathbf{x}$) are used for vectors, but only where the fact that a variable is a vector is relevant. Square brackets indicate elements of a matrix or vector: if $\mathbf{x} = [x_1, \dots, x_N]$ is a row vector, then $[x]_i = x_i$ denotes its entries; if $A \in \mathbb{R}^{n \times m}$ is a matrix, then $[A]_{ij} = A_{ij}$ denotes its entries. Round brackets $(\cdot)$ are used in most other cases (as in the notations listed below).

Notation	Meaning
$a \propto c$	a is proportional to c : there is a constant k such that $a = k \cdot c$.
$A \wedge B, A \vee B$	The logical conjunctions “and” and “or”; i.e. $A \wedge B$ is true iff both A and B are true, $A \vee B$ is true iff $\neg A \wedge \neg B$ is false.
$A \otimes B$	The Kronecker product of matrices A, B . See Eq. (15.2).
$A \circledast B$	The symmetric Kronecker product . See Eq. (19.16).
$A \odot B$	The element-wise product (aka Hadamard product) of two matrices A and B of the same shape, i.e. $[A \odot B]_{ij} = [A]_{ij} \cdot [B]_{ij}$.
$\vec{A}, \mathbb{A}$	$\vec{A}$ is the vector arising from stacking the elements of a matrix A row after row, and its inverse ($A = \mathbb{A}\vec{A}$). See Eq. (15.1).
$\text{cov}_p(x, y)$	The covariance of x and y under p . That is, $\text{cov}_p(x, y) := \mathbb{E}_p(x \cdot y) - \mathbb{E}_p(x)\mathbb{E}_p(y)$.
$C^q(V, \mathbb{R}^d)$	The set of q -times continuously differentiable functions from V to $\mathbb{R}^d$, for some $q, d \in \mathbb{N}$.
$\delta(x - y)$	The Dirac delta , heuristically characterised by the property $\int f(x)\delta(x - y) dx = f(y)$ for functions $f : \mathbb{R} \rightarrow \mathbb{R}$.
δ_{ij}	The Kronecker symbol : $\delta_{ij} = 1$ if $i = j$, otherwise $\delta_{ij} = 0$.
$\det(A)$	The determinant of a square matrix A .
$\text{diag}(\mathbf{x})$	The diagonal matrix with entries $[\text{diag}(\mathbf{x})]_{ij} = \delta_{ij}x_i$.
$d\omega_t$	The notation for an Itô integral in a stochastic differential equation . See Definition 5.4.
$\text{erf}(x)$	The error function $\text{erf}(x) := \frac{2}{\sqrt{\pi}} \int_0^x \exp(-t^2) dt$.
$\mathbb{E}_p(f)$	The expectation of f under p . That is, $\mathbb{E}_p(f) := \int f(x) dp(x)$.
$\mathbb{E}_{ Y}(f)$	The expectation of f under $p(f Y)$.
$\Gamma(z)$	The Gamma function $\Gamma(z) := \int_0^\infty x^{z-1} \exp(-x) dx$. See Eq. (6.1).
$\mathcal{G}(\cdot; a, b)$	The Gamma distribution with shape $a > 0$ and rate $b > 0$, with probability density function $\mathcal{G}(z; a, b) := \frac{b^a z^{a-1}}{\Gamma(a)} e^{-bz}$.
$\mathcal{GP}(f; \mu, k)$	The Gaussian process measure on f with mean function μ and covariance function (kernel) k . See §4.2
$\mathbb{H}_p(x)$	The (differential) entropy of the distribution $p(x)$. That is, $\mathbb{H}_p(x) := - \int p(x) \log p(x) dx$. See Eq. (3.2).
$\mathbb{H}(x y)$	The (differential) entropy of the cond. distribution $p(x y)$. That is, $\mathbb{H}(x y) := \mathbb{H}_{p(\cdot y)}(x)$.
$I(x; y)$	The mutual information between random variables X and Y . That is, $I(x; y) := \mathbb{H}(x) - \mathbb{H}(x y) = \mathbb{H}(y) - \mathbb{H}(y x)$.

Notation	Meaning
I, I_N	The identity matrix (of dimensionality N): $[I]_{ij} = \delta_{ij}$.
$\mathbb{I}(\cdot \in A)$	The indicator function of a set A .
K_ν	The modified Bessel function for some parameter $\nu \in \mathbb{C}$. That is, $K_\nu(x) := \int_0^\infty \exp(-x \cdot \cosh(t)) \cosh(\nu t) dt$.
$\mathcal{L}$	The loss function of an optimization problem (§26.1), or the log-likelihood of an inverse problem (§41.2).
$\mathcal{M}$	The model $\mathcal{M}$ capturing the probabilistic relationship between the latent object and computable quantities. See §9.3.
$\mathbb{N}, \mathbb{C}, \mathbb{R}, \mathbb{R}_+$	The natural numbers (excluding zero), the complex numbers, the real numbers, and the positive real numbers, respectively.
$\mathcal{N}(x; \mu, \Sigma) = p(x)$	The vector x has the Gaussian probability density function with mean vector μ and covariance matrix Σ . See Eq. (3.1).
$\mathcal{N}(\mu, \Sigma) \sim X$	The random variable X is distributed according to a Gaussian distribution with mean μ and covariance Σ .
$\mathcal{O}(\cdot)$	Landau big-Oh : for functions f, g defined on $\mathbb{N}$, the notation $f(n) = \mathcal{O}(g(n))$ means that $f(n)/g(n)$ is bounded for $n \rightarrow \infty$.
$p(y x)$	The conditional the probability density function for variable Y having value y conditioned on variable X having value x .
$\text{rk}(A)$	The rank of a matrix A .
$\text{span}\{x_1, \dots, x_n\}$	The linear span of $\{x_1, \dots, x_n\}$.
$\text{St}(\cdot; \mu, \lambda_1, \lambda_1)$	The Student's-t probability density function with parameters $\mu \in \mathbb{R}$ and $\lambda_1, \lambda_2 > 0$, see Eq. (6.9).
$\text{tr}(A)$	The trace of matrix A . That is, $\text{tr}(A) = \sum_i [A]_{ii}$.
$A^\top$	The transpose of matrix A : $[A^\top]_{ij} = [A]_{ji}$.
$\mathbb{U}_{a,b}$	The uniform distribution with probability density function $p(u) := \mathbb{I}(u \in (a, b))$, for $a < b$.
$\mathbb{V}_p(x)$	The variance of x under p . That is, $\mathbb{V}_p(x) := \text{cov}_p(x, x)$.
$\mathbb{V}_Y(f)$	The variance of f under $p(f Y)$. That is, $\mathbb{H}(x y) := - \int \log p(x y) dp(x y)$.
$\mathcal{W}(V, \nu)$	The Wishart distribution with probability density function $\mathcal{W}(x; V, \nu) \propto x ^{(\nu-N-1)/2} e^{-1/2 \text{tr}(V^{-1}x)}$. See Eq. (19.1).
$x \perp y$	x is orthogonal to y , i.e. $\langle x, y \rangle = 0$.
$x := a$	The object x is defined to be equal to a .
$x \stackrel{\Delta}{=} a$	The object x is equal to a by virtue of its definition.
$x \leftarrow a$	The object x is assigned the value of a (used in pseudo-code).
$X \sim p$	The random variable X is distributed according to p .
$\mathbf{1}, \mathbf{1}_d$	A column vector of d ones , $\mathbf{1}_d := [1, \dots, 1]^\top \in \mathbb{R}^d$.
$\nabla_x f(x, t)$	The gradient of f w.r.t. x . (We omit subscript x if redundant.)